

The Problem of Local Extrema

Every algorithm that tries to find a maximum or minimum faces a very serious potential problem with local extrema (local maximums and minimums). We may perform gradient ascent perfectly, but realize that the peak we have reached at the end is only a “local” peak—it’s higher than every point around it, but not higher than some faraway global maximum. This could happen in real life as well: you try to climb a mountain, you reach a summit where you are higher than all of your immediate surroundings, but you realize that you’re only on the foothill and the real summit is far away and much higher. Paradoxically, you may have to walk down a little to eventually get to that higher summit, so the “naive” strategy that gradient ascent follows, always stepping to a slightly higher point in one’s immediate neighborhood, fails to get to the global maximum.

Education and Lifetime Income

Local extrema are a very serious problem in gradient ascent. As an example, consider trying to maximize lifelong income by choosing the optimal level of education. In this case, we might suppose that lifelong earnings relate to years of education according to the following formula:

```
import math
def income(edu_yrs):
    return(math.sin((edu_yrs - 10.6) * (2 * math.pi/4)) + (edu_yrs - 11)/2)
```

Here, `edu_yrs` is a variable expressing how many years of education one has received, and `income` is a measurement of one’s lifetime income. We can plot this curve as follows, including a point for a person who has 12.5 years of formal education—that is, someone who has graduated from high school (12 years of formal education) and is half a year into a bachelor’s degree program:

```
import matplotlib.pyplot as plt
xs = [11 + x/100 for x in list(range(901))]
ys = [income(x) for x in xs]
plt.plot(xs,ys)
current_edu = 12.5
plt.plot(current_edu,income(current_edu),'ro')
plt.title('Education and Income')
```
